we considered the paper technically sound and a borderline accept for the workshop, acknowledging its valuable insights and intriguing ideas. However, we concluded it lacks sufficient depth and rigor for acceptance into a full conference without addressing the highlighted concerns.

**Paper Assessment by Human Workshop Reviewers.** The reviewers generally agree that the paper addresses an important topic—compositional generalization in neural networks and appreciate the authors' proposed compositional regularization method, as well as their detailed analysis of unexpected results. All reviewers recognize the paper's strength in clearly presenting why the regularization term does not yield the anticipated improvements, emphasizing its informative negative results. However, the reviewers highlight several areas for improvement:

*Justification and Intuition*: All reviewers suggest the need for clearer justification or intuition behind why penalizing large changes between successive hidden states might improve compositionality. They recommend adding references to related works, theoretical motivations, or visual explanations to strengthen the rationale.
*Network Architecture Generalization*: Reviewers emphasize that since only the LSTM architecture was evaluated, the findings should not be generalized across all neural network types. They suggest experimenting with additional architectures, such as transformers, to better understand the impact of the regularization across different neural network models.
*Experimental Breadth*: Reviewers suggest extending the evaluation to other tasks or datasets beyond synthetic arithmetic expressions to further validate the generalizability of the conclusions.
*Overall*: The reviewers recommend acceptance to the workshop due to the paper's insightful exploration and clear analysis despite its negative results. They encourage further elaboration on methodological motivations, additional experimental evaluations, and clearer connections between compositional regularization and the complexity of compositional tasks. The paper received scores of 6 (weak accept), 7 (accept), and 6 (weak accept). Below, we include two of the reviews for which we obtained explicit permission from the reviewers to include them in our report. The remaining reviewer did not respond to our request.

**Reviewer #1: A good paper analyzing the effectiveness of a compositional regularization term for LSTMs**

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Summary: The authors propose a regularisation term to enhance compositional regularisation in
neural networks. The idea is to penalise large deviations between subsequent time steps of the
hidden state, thus "squeezing" the hidden state to encourage composition and preventing a
dominating representation. The authors test their approach on synthetic arithmetic expression
with varying operator complexity and length. They show that although the regularisation term
appears to be working, it counterintuitively does not improve test accuracy. Furthermore, the
authors identify a bottleneck regarding network capacity with increasing arithmetic operators.

Strengths:
I find the idea of regularising or squeezing the hidden representations to encourage
compositionally an interesting idea. The authors define a good baseline and ablate their method
well against it, revealing why the regularisation term does not work as expected. I think the
insight that operator complexity is a bottleneck for the neural network is important, as it
raises the question whether architectural changes might be more effective for compositionally
than regularisation.

Weaknesses:
The paper would benefit from more intuition as to why the proposed regularisation term should
encourage compositionality. This could be either an experiment or simply a visualisation for the
reader. Only one architecture (LSTM) was tested. It would be interesting to see if transformer
architectures fare better with compositionality due to the attention mechanism.
I think the connection between compositional regularisation and operator complexity needs to be
made more explicit. From reading the introduction both arguments seem a bit disconnected although
I can infer the authors intentions.

Conclusion:
Overall, I would accept this paper to the workshop, since it proposes a simple and interesting
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